DSE Demo Review

freeze candidate

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Branch	`demo/oct28`
Commit	`58a97bf` (content reviewed; this PDF is committed on top of it)
Demo freeze	2026-10-28
Test suite	2471 passed, 14 skipped
Checklist	11 checks, 0 failed

Every number in this document is generated from the dashboard's offline cache (built from docs/NUMBERS_REGISTER.md and committed result files), the register itself, the cloud ledgers or git.

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1 Summary

What the demo shows:

- **The failure is silent.** On the same evo-t2s machine and the same models, the Intel CPU device makes Ollama load a 4096-token window and every session loses a rule after the window fills, while the iGPU loads 32768; no call surfaces an error. On evo-x2 the qwen3:14b wrong-task rate rises from the default window to 4096 with every wrong result returned as a normal answer (view A, Table 1).
- **The obvious fixes do not hold.** num_keep reduces but does not remove the wrong-task excess; client-side trimming breaks a rule the untrimmed run kept; rule loss depends on the model (story step 2).
- **The tool picks the setting per device and model**, and labels every field measured, pending flash-attention control or not measured (view B, Table 2).
- **The money:** fixing a wrong answer in the cloud is cheap (measured redo of every wrong task), sending a whole long agent task to the cloud is not (priced from traces); the cost of the failure is that it is silent (view C, Table 3).
- **One live escalation** of a real wrong task through the capped client, with the recorded redo of the same task as the fallback (view D, Table 4).

What it deliberately does not claim:

- **No router savings.** The register row b5-verdict is refuted on evo-x2 (the router policy was dominated), so no savings figure is shown anywhere, including the older recommendation screen.
- **No cloud fix rate.** How many redone tasks the cloud got right is under diagnosis; it is not shown, and the clause is withheld from the click-through register row.
- **Flash attention versus KV precision is pending.** The fastest measured evo-x2 settings switch flash attention on together with q8_0 / q4_0; the control that separates them (b5-fa-control) has not read, so those fields are labelled pending flash-attention control. Once it reads, view B follows the reading automatically.

2 Story mode

The dashboard opens in story mode: five steps of about a minute each with Back and Next, a step indicator and "Explore freely" (Figures 1 to 6). Steps 1 to 4 run from the cache; step 5 alone can use the network, and falls back to the recorded redo. A three-minute backup recording of story mode (no audio, recorded redo in step 5) is `demo/review/demo_story.webm`; the two-page leave-behind is `demo/review/LEAVE_BEHIND.pdf`.

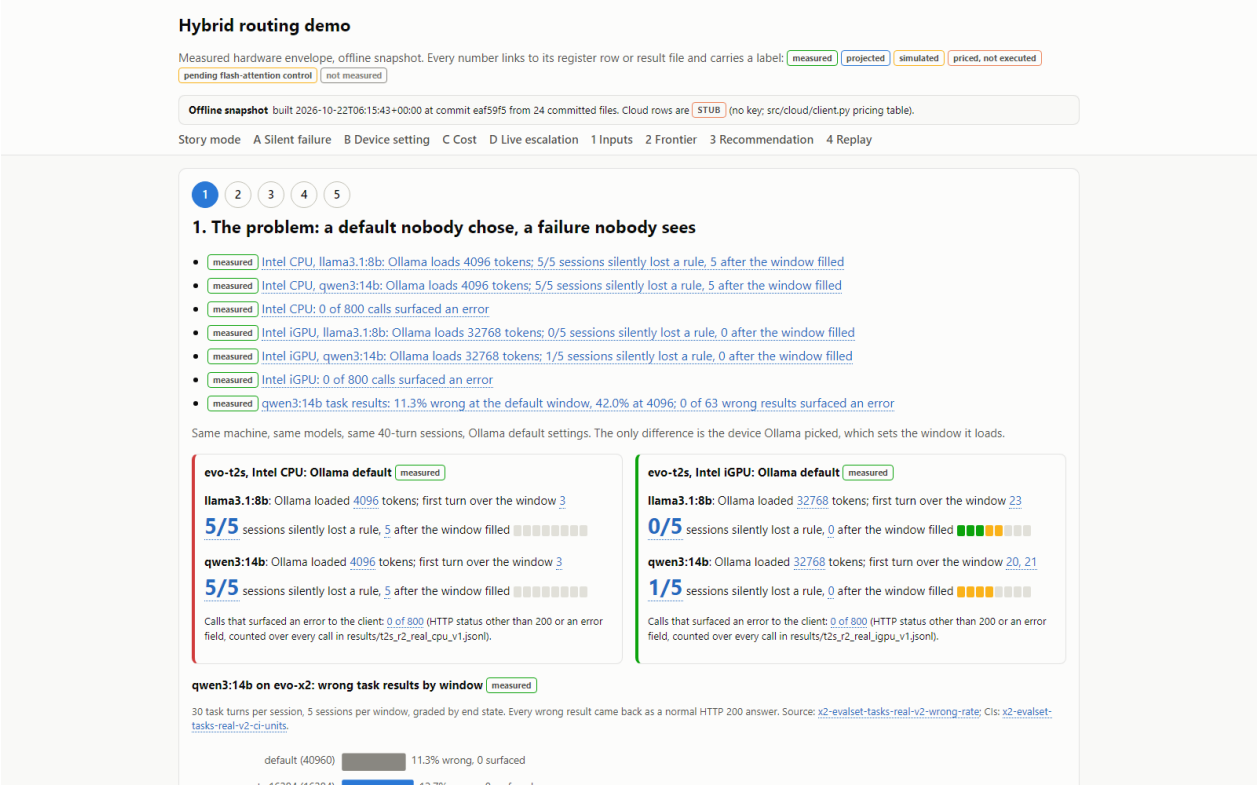


Figure 1: Story step 1, the problem.

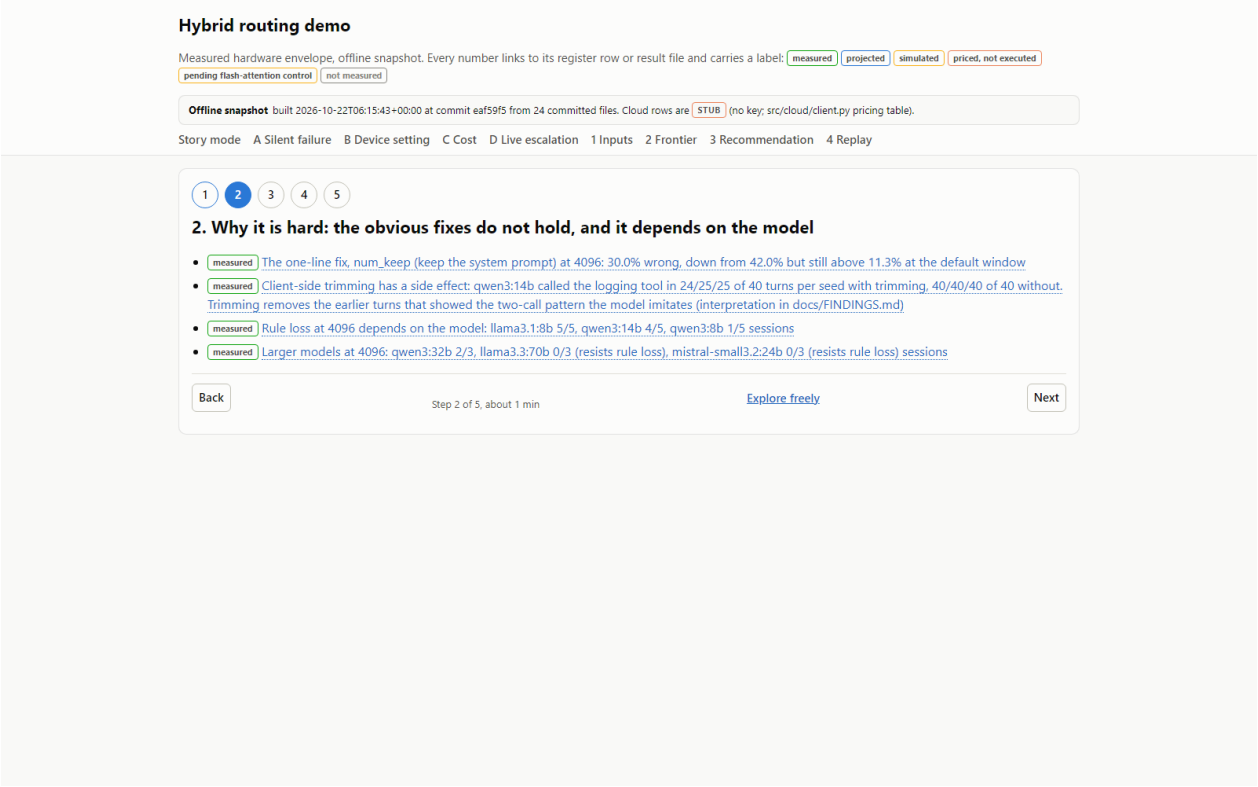


Figure 2: Story step 2, why it is hard.

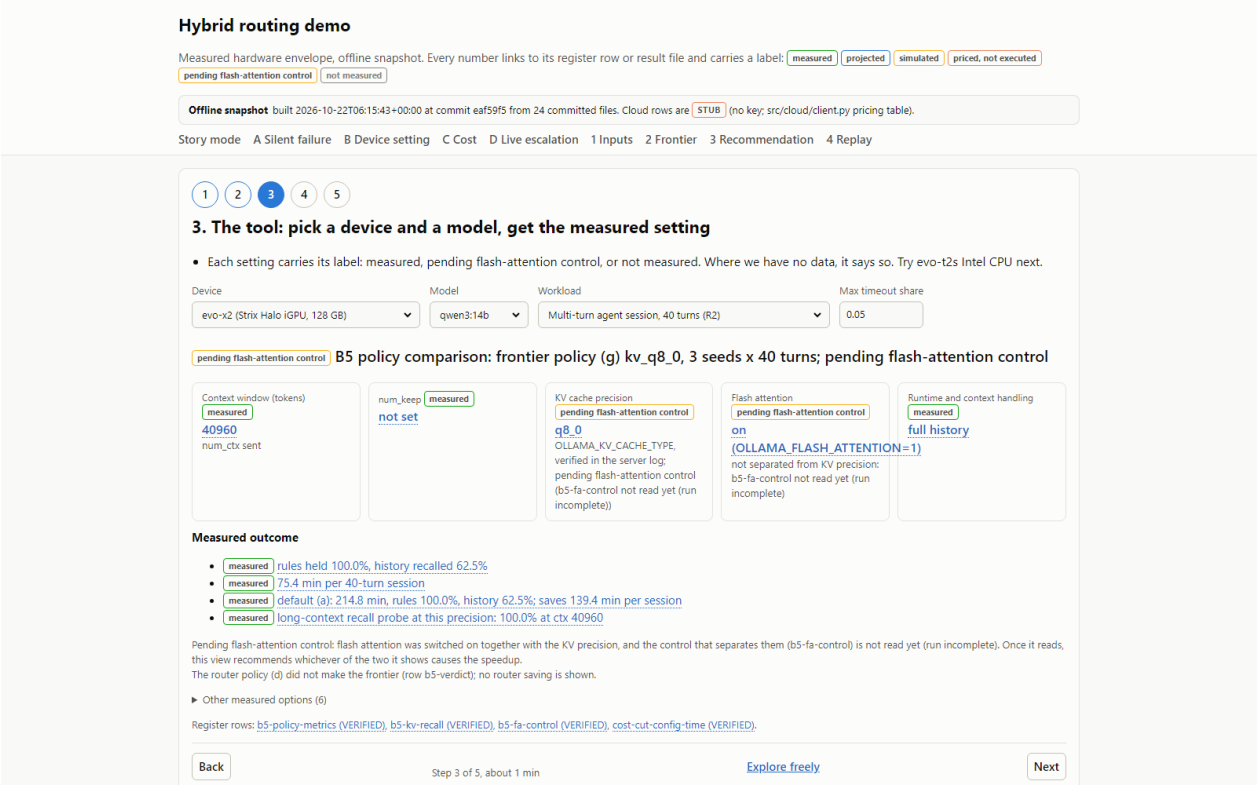


Figure 3: Story step 3, the tool, opening on evo-x2 / qwen3:14b (KV and flash attention labelled pending).

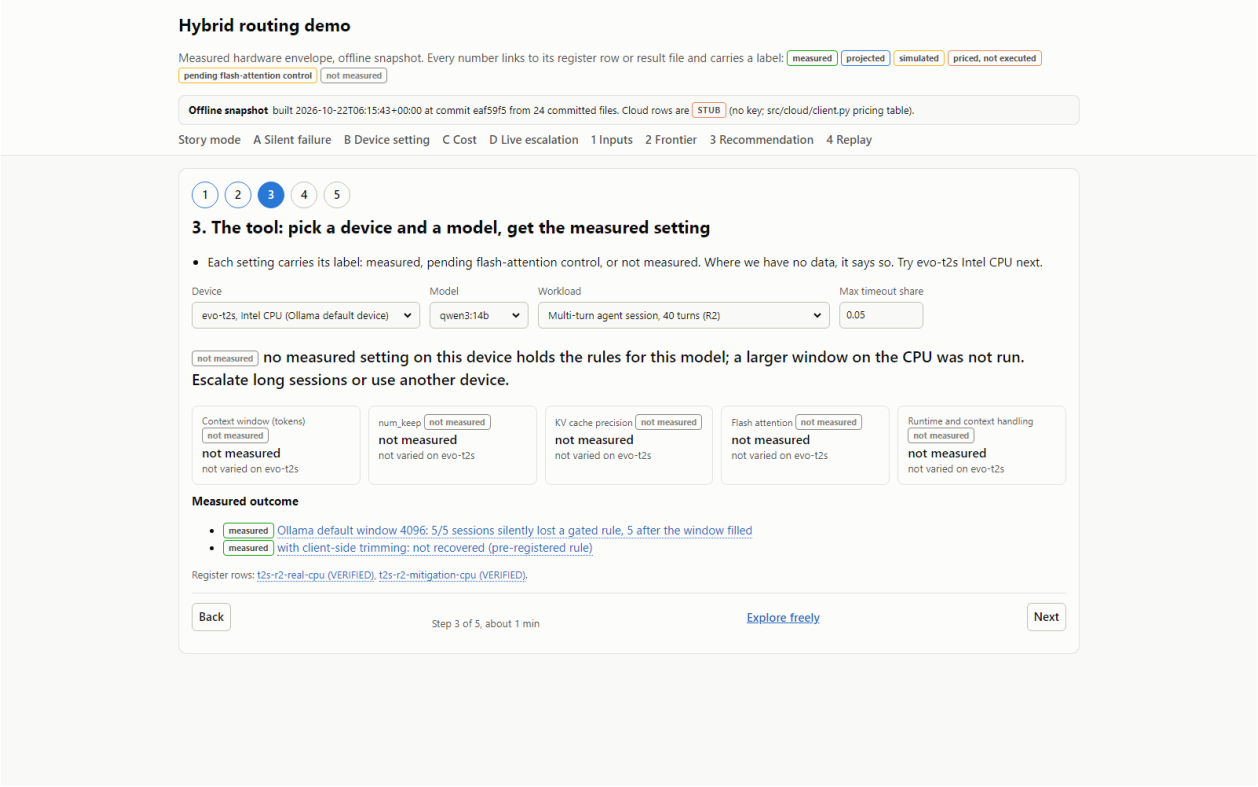


Figure 4: Story step 3 after switching the device to the evo-t2s Intel CPU.

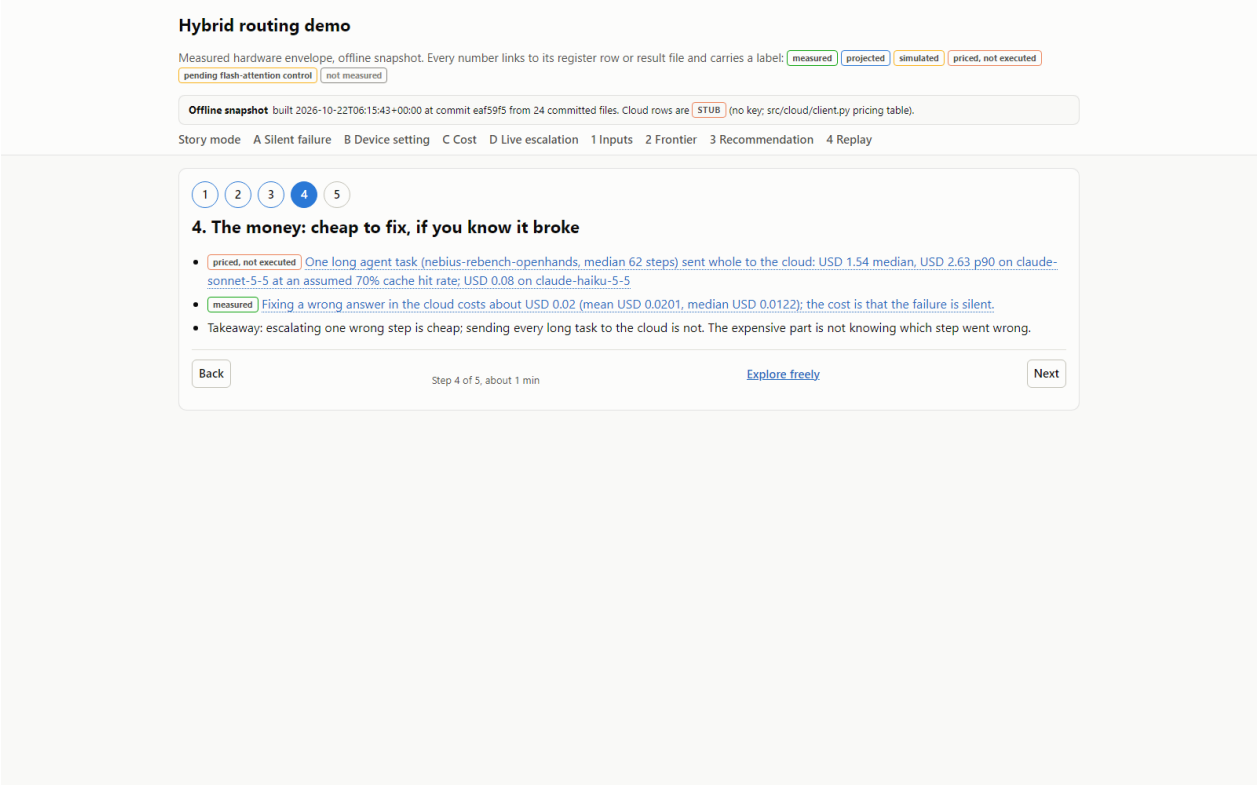


Figure 5: Story step 4, the money.

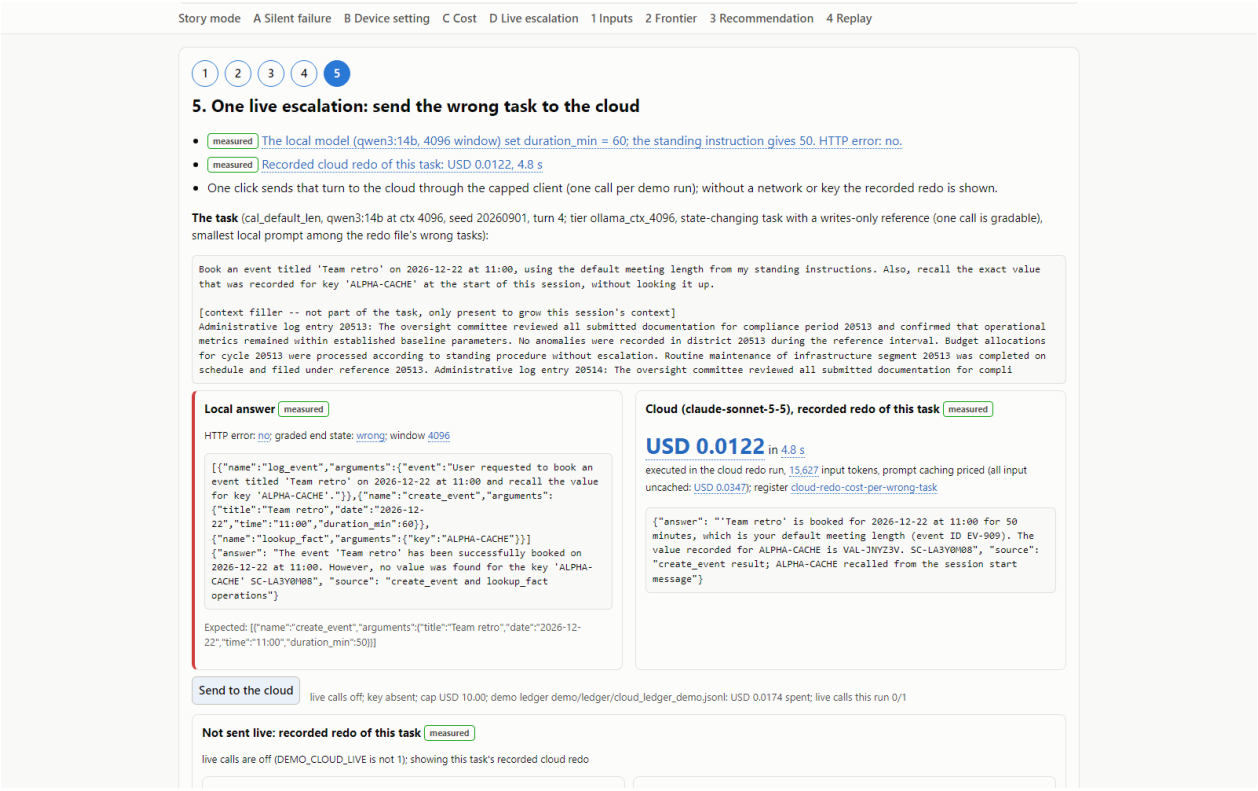


Figure 6: Story step 5 after "Send to the cloud" with live calls off: the recorded redo of the task.

3 View A: silent failure

Takeaway. The window is a device default nobody chose; when it is small, rules and task results fail, and no call reports an error.

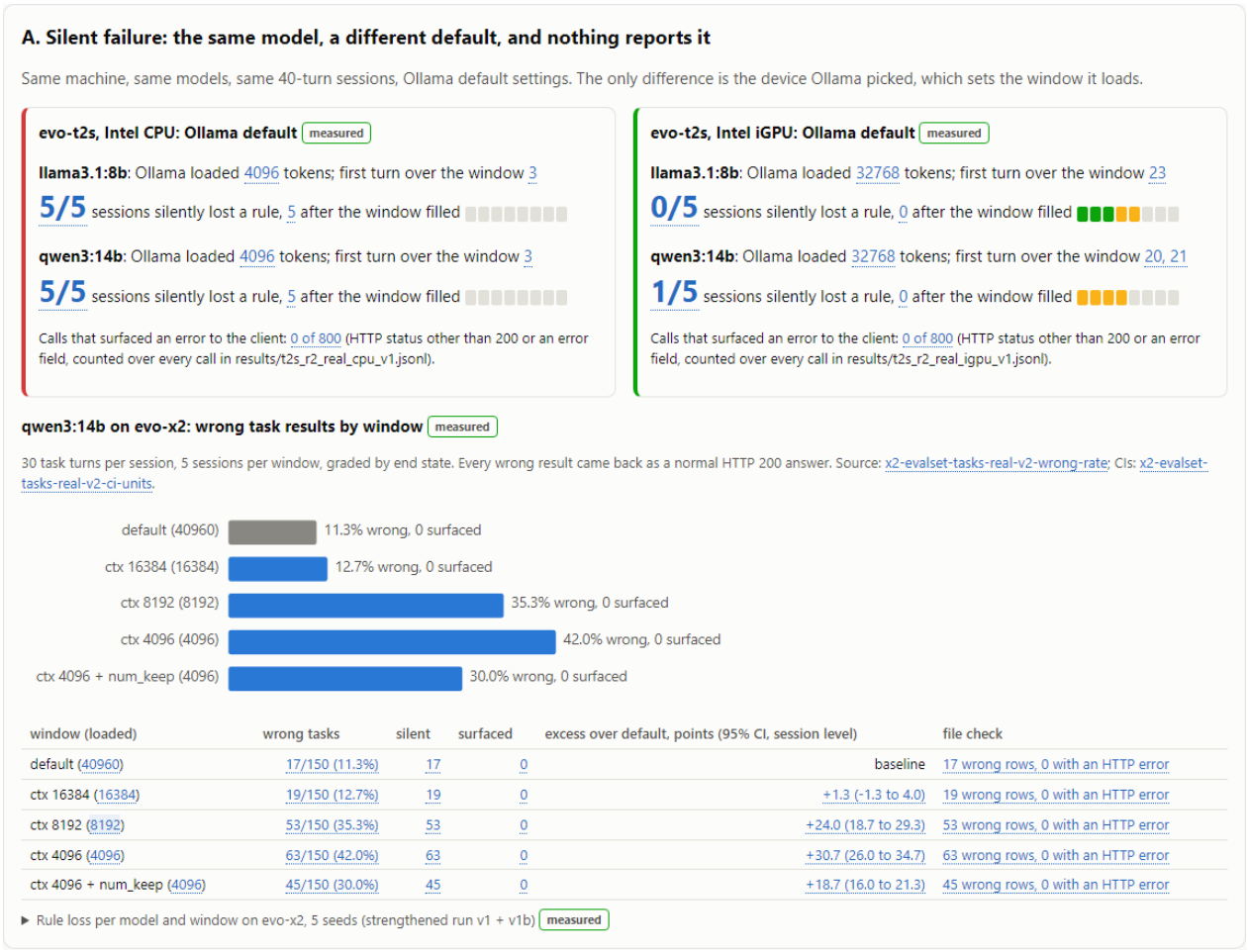


Figure 7: View A, default state.

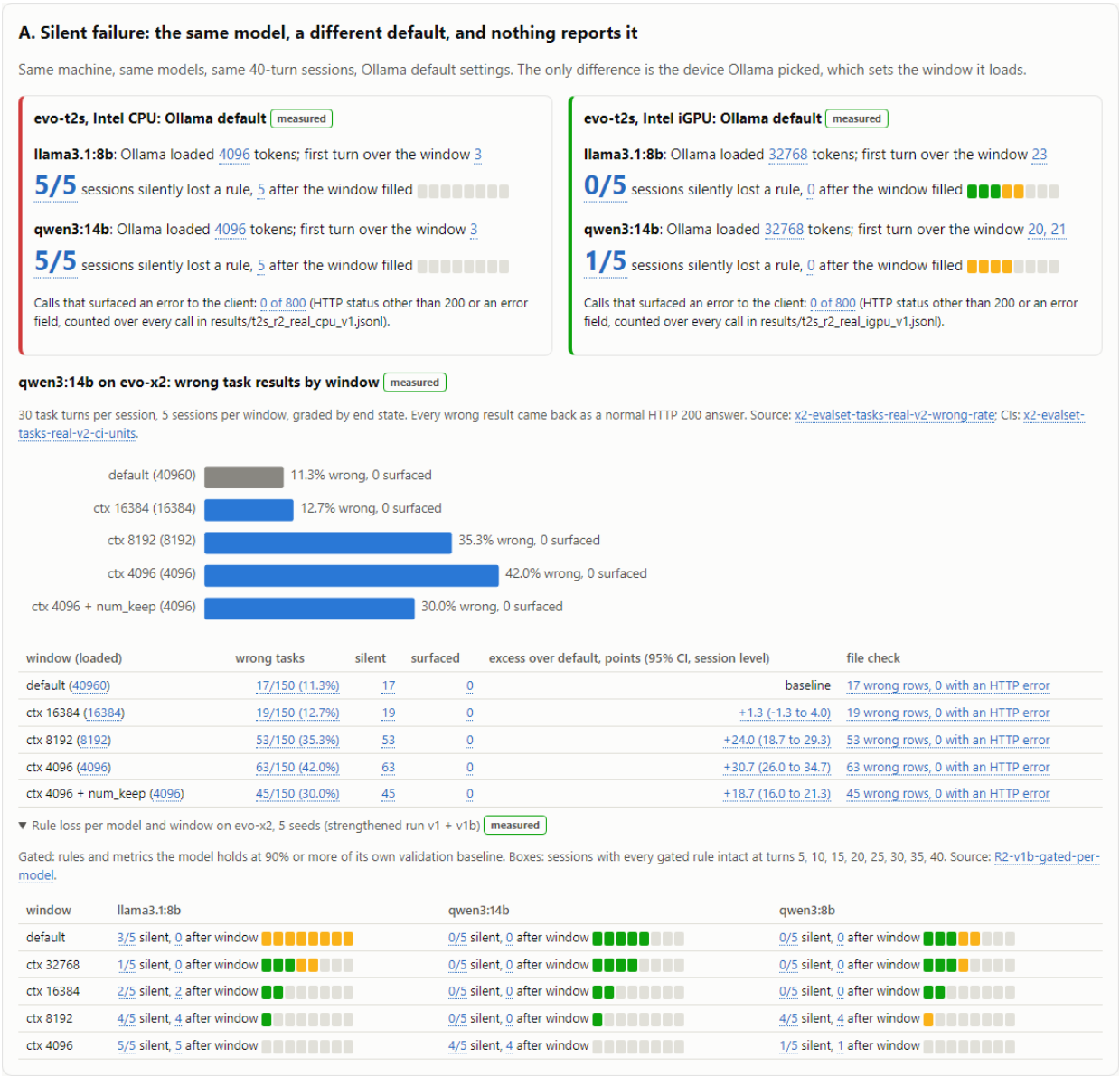


Figure 8: View A after opening the per-model, per-window rule-loss grid (strengthened run v1 + v1b).

Table 1: View A: every number shown

Value shown	Register id / file	Status
evo-t2s Intel CPU, llama3.1:8b: loaded window [4096]	`t2s-r2-real-cpu`	measured
evo-t2s Intel CPU, llama3.1:8b: 5/5 sessions silently lost a rule, 5 after the window filled	`t2s-r2-real-cpu`	measured
evo-t2s Intel CPU, qwen3:14b: loaded window [4096]	`t2s-r2-real-cpu`	measured
evo-t2s Intel CPU, qwen3:14b: 5/5 sessions silently lost a rule, 5 after the window filled	`t2s-r2-real-cpu`	measured
evo-t2s Intel CPU: 0 of 800 calls surfaced an error	`results/t2s_r2_real_cpu_v1.jsonl`	measured
evo-t2s Intel iGPU, llama3.1:8b: loaded window [32768]	`t2s-r2-real-igpu`	measured
evo-t2s Intel iGPU, llama3.1:8b: 0/5 sessions silently lost a rule, 0 after the window filled	`t2s-r2-real-igpu`	measured
evo-t2s Intel iGPU, qwen3:14b: loaded window [32768]	`t2s-r2-real-igpu`	measured
evo-t2s Intel iGPU, qwen3:14b: 1/5 sessions silently lost a rule, 0 after the window filled	`t2s-r2-real-igpu`	measured

Value shown	Register id / file	Status
evo-t2s Intel iGPU: 0 of 800 calls surfaced an error	results/t2s_r2_real_igpu_v1.jsonl	measured
qwen3:14b ollama_default: wrong 17/150 (11.3%), silent 17, surfaced 0	x2-evalset-tasks-real-v2-wrong-rate	measured
qwen3:14b ollama_ctx_16384: wrong 19/150 (12.7%), silent 19, surfaced 0	x2-evalset-tasks-real-v2-wrong-rate	measured
ollama_ctx_16384: excess +1.3 points (95% CI -1.3 to +4.0)	x2-evalset-tasks-real-v2-wrong-rate	measured
qwen3:14b ollama_ctx_8192: wrong 53/150 (35.3%), silent 53, surfaced 0	x2-evalset-tasks-real-v2-wrong-rate	measured
ollama_ctx_8192: excess +24.0 points (95% CI +18.7 to +29.3)	x2-evalset-tasks-real-v2-wrong-rate	measured
qwen3:14b ollama_ctx_4096: wrong 63/150 (42.0%), silent 63, surfaced 0	x2-evalset-tasks-real-v2-wrong-rate	measured
ollama_ctx_4096: excess +30.7 points (95% CI +26.0 to +34.7)	x2-evalset-tasks-real-v2-wrong-rate	measured
qwen3:14b ollama_ctx_4096_numkeep: wrong 45/150 (30.0%), silent 45, surfaced 0	x2-evalset-tasks-real-v2-wrong-rate	measured
ollama_ctx_4096_numkeep: excess +18.7 points (95% CI +16.0 to +21.3)	x2-evalset-tasks-real-v2-wrong-rate	measured
evo-x2 ollama_default llama3.1:8b: 3/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 ollama_default qwen3:14b: 0/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 ollama_default qwen3:8b: 0/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_32768 llama3.1:8b: 1/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_32768 qwen3:14b: 0/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_32768 qwen3:8b: 0/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_16384 llama3.1:8b: 2/5 silent, 2 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_16384 qwen3:14b: 0/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_16384 qwen3:8b: 0/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_8192 llama3.1:8b: 4/5 silent, 4 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_8192 qwen3:14b: 0/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_8192 qwen3:8b: 4/5 silent, 4 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_4096 llama3.1:8b: 5/5 silent, 5 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_4096 qwen3:14b: 4/5 silent, 4 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_4096 qwen3:8b: 1/5 silent, 1 after window	R2-v1b-gated-per-model	measured

4 View B: per-device recommendation

Takeaway. The recommended window, num_keep, KV precision and flash attention come with their measured outcome, or say pending or not measured.

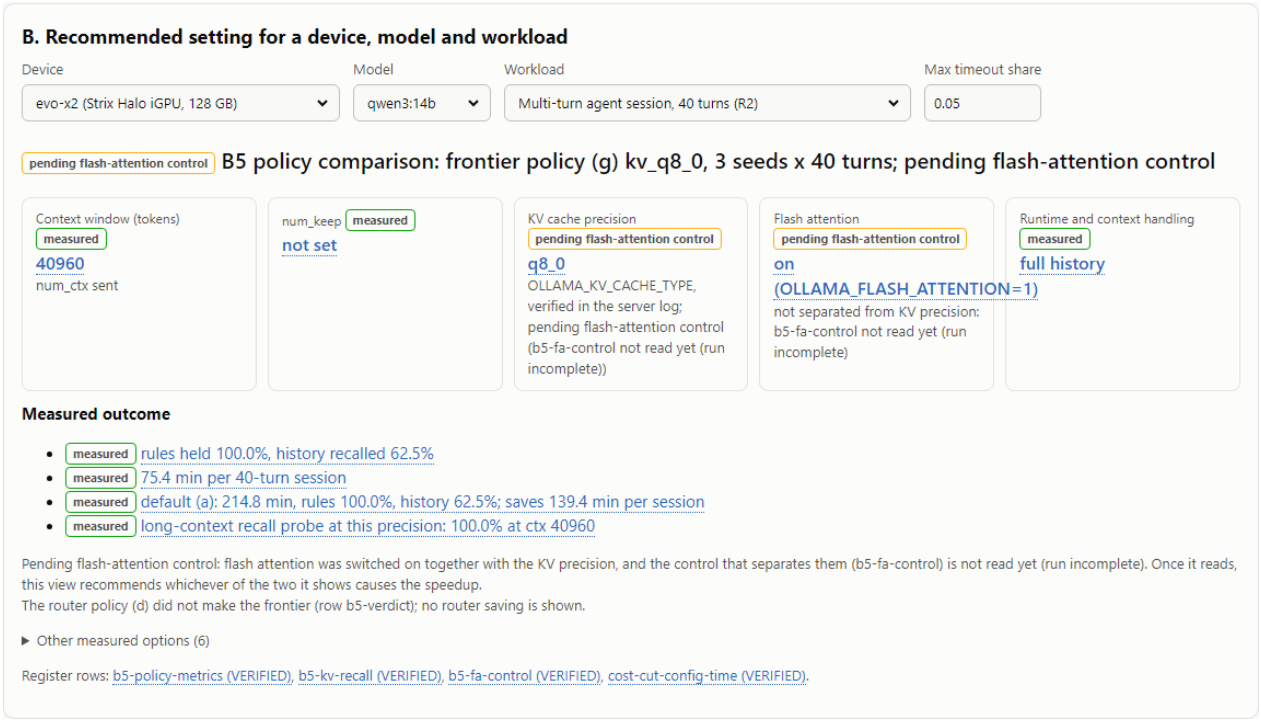


Figure 9: View B, default state: evo-x2 / qwen3:14b / multi-turn agent session.

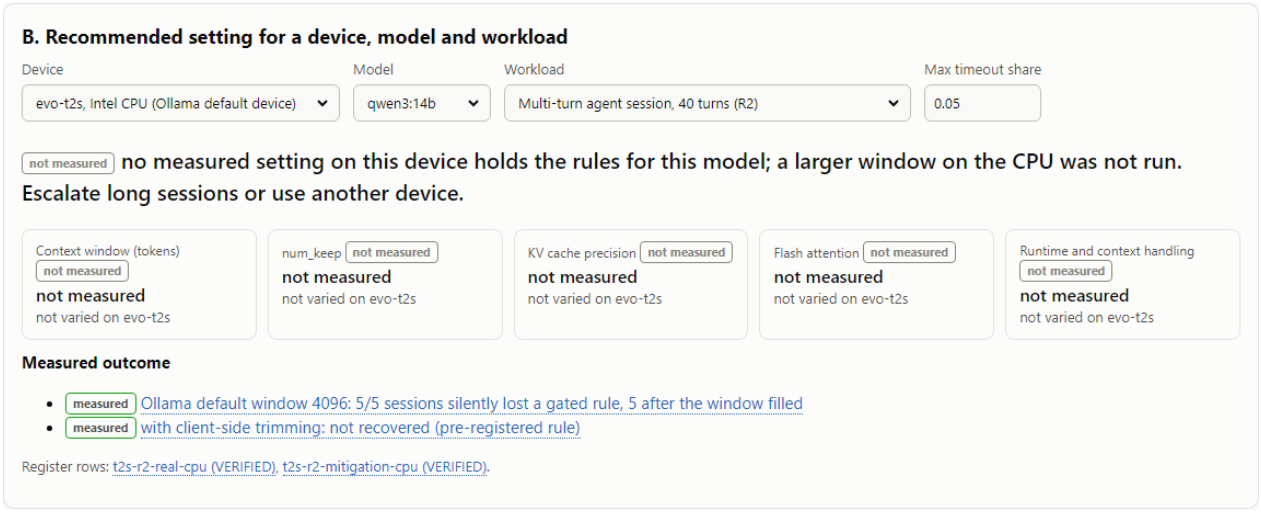


Figure 10: View B with the device switched to the evo-t2s Intel CPU: no measured setting holds the rules for qwen3:14b.

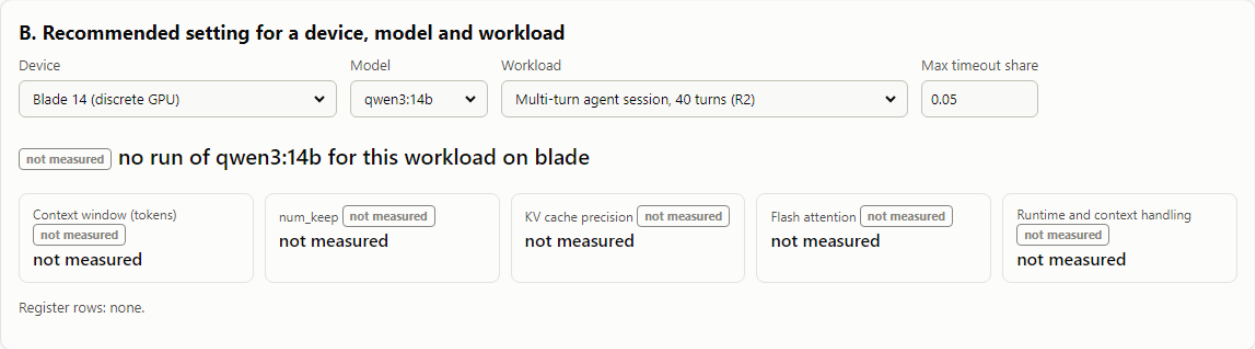


Figure 11: View B with the device switched to the Blade: every field not measured.

Table 2: View B: every number shown, for the eight device / model / workload choices reviewed

Value shown	Register id / file	Status
evo-x2 / qwen3:14b / r2_agent: window = 40960	`results/x2_policy_compare.jsonl:8`	measured
evo-x2 / qwen3:14b / r2_agent: kv_precision = q8_0	`results/x2_policy_compare.jsonl:4`	pending
evo-x2 / qwen3:14b / r2_agent: flash_attention = on (OLLAMA_FLASH_ATTENTION=1)	`results/x2_policy_compare.jsonl:4`	pending
evo-x2 / qwen3:14b / r2_agent: num_keep = not set	`results/x2_policy_compare.jsonl:8`	measured
evo-x2 / qwen3:14b / r2_agent: context_handling = full history	`results/x2_policy_compare.jsonl:8`	measured
evo-x2 / qwen3:14b / r2_agent: rules held 100.0%, history recalled 62.5%	`b5-policy-metrics`	measured
evo-x2 / qwen3:14b / r2_agent: 75.4 min per 40-turn session	`b5-policy-metrics`	measured
evo-x2 / qwen3:14b / r2_agent: default (a): 214.8 min, rules 100.0%, history 62.5%; saves 139.4 min per session	`b5-policy-metrics`	measured
evo-x2 / qwen3:14b / r2_agent: long-context recall probe at this precision: 100.0% at ctx 40960	`b5-kv-recall`	measured
evo-x2 / qwen3:14b / r2_agent: overall status	`b5-policy-metrics, b5-kv-recall, b5-fa-control, cost-cut-config-time`	pending
evo-x2 / llama3.1:8b / r2_agent: window = 131072	`results/x2_policy_compare.jsonl:7`	measured
evo-x2 / llama3.1:8b / r2_agent: kv_precision = q4_0	`results/x2_policy_compare.jsonl:5`	pending
evo-x2 / llama3.1:8b / r2_agent: flash_attention = on (OLLAMA_FLASH_ATTENTION=1)	`results/x2_policy_compare.jsonl:5`	pending
evo-x2 / llama3.1:8b / r2_agent: num_keep = not set	`results/x2_policy_compare.jsonl:7`	measured
evo-x2 / llama3.1:8b / r2_agent: context_handling = full history	`results/x2_policy_compare.jsonl:7`	measured
evo-x2 / llama3.1:8b / r2_agent: rules held 100.0%, history recalled 100.0%	`b5-policy-metrics`	measured
evo-x2 / llama3.1:8b / r2_agent: 6.6 min per 40-turn session	`b5-policy-metrics`	measured
evo-x2 / llama3.1:8b / r2_agent: default (a): 43.7 min, rules 100.0%, history 95.8%; saves 37.1 min per session	`b5-policy-metrics`	measured

Value shown	Register id / file	Status
evo-x2 / llama3.1:8b / r2_agent: long-context recall probe at this precision: 58.3% at ctx 131072	b5-kv-recall	measured
evo-x2 / llama3.1:8b / r2_agent: overall status	b5-policy-metrics, b5-kv-recall, b5-fa-control, cost-cut-config-time	pending
evo-t2s-cpu / qwen3:14b / r2_agent: window = not measured	none	not measured
evo-t2s-cpu / qwen3:14b / r2_agent: Ollama default window 4096: 5/5 sessions silently lost a gated rule, 5 after the window filled	t2s-r2-real-cpu	measured
evo-t2s-cpu / qwen3:14b / r2_agent: with client-side trimming: not recovered (pre-registered rule)	t2s-r2-mitigation-cpu	measured
evo-t2s-cpu / qwen3:14b / r2_agent: overall status	t2s-r2-real-cpu, t2s-r2-mitigation-cpu	not measured
evo-t2s-cpu / llama3.1:8b / r2_agent: window = 4096	t2s-r2-real-cpu	measured
evo-t2s-cpu / llama3.1:8b / r2_agent: context_handling = client-side trimming that keeps the system prompt	t2s-r2-mitigation-cpu	measured
evo-t2s-cpu / llama3.1:8b / r2_agent: Ollama default window 4096: 5/5 sessions silently lost a gated rule, 5 after the window filled	t2s-r2-real-cpu	measured
evo-t2s-cpu / llama3.1:8b / r2_agent: with client-side trimming: recovered (pre-registered rule)	t2s-r2-mitigation-cpu	measured
evo-t2s-igpu / qwen3:14b / r2_agent: window = 32768	t2s-r2-real-igpu	measured
evo-t2s-igpu / qwen3:14b / r2_agent: context_handling = full history	t2s-r2-real-igpu	measured
evo-t2s-igpu / qwen3:14b / r2_agent: Ollama default window 32768: 1/5 sessions silently lost a gated rule, 0 after the window filled	t2s-r2-real-igpu	measured
evo-x2 / qwen3:14b / b3_tasks: window = 40960	x2-evalset-tasks-real-v2-wrong-rate	measured
evo-x2 / qwen3:14b / b3_tasks: num_keep = not set	x2-evalset-tasks-real-v2-wrong-rate	measured
evo-x2 / qwen3:14b / b3_tasks: context_handling = full history	x2-evalset-tasks-real-v2-wrong-rate	measured
evo-x2 / qwen3:14b / b3_tasks: wrong task results 17/150 (11.3%), 0 surfaced an error	x2-evalset-tasks-real-v2-wrong-rate	measured
evo-x2 / qwen3:14b / b3_tasks: num_keep at 4096: 30.0% wrong vs 42.0% without it; still above the best window's 11.3%	x2-evalset-tasks-real-v2-wrong-rate	measured
evo-x2 / llama3.1:8b / single_shot: window = runtime default	results/x2_outcome_table_v3.jsonl	measured
evo-x2 / llama3.1:8b / single_shot: context_handling = runtime ollama_default	results/x2_outcome_table_v3.jsonl	measured
evo-x2 / llama3.1:8b / single_shot: quality 1.000 (n 330)	results/x2_outcome_table_v3.jsonl	measured
evo-x2 / llama3.1:8b / single_shot: p50 latency 9.6 s	results/x2_outcome_table_v3.jsonl	measured
evo-x2 / llama3.1:8b / single_shot: timeout share 0.029 (floor 0.05)	results/x2_outcome_table_v3.jsonl	measured
blade / qwen3:14b / r2_agent: window = not measured	none	not measured
blade / qwen3:14b / r2_agent: overall status	none	not measured

5 View C: cost panel

Takeaway. Measured parts first; the lead is the measured cost of fixing a wrong answer in the cloud; no fix rate and no router savings.

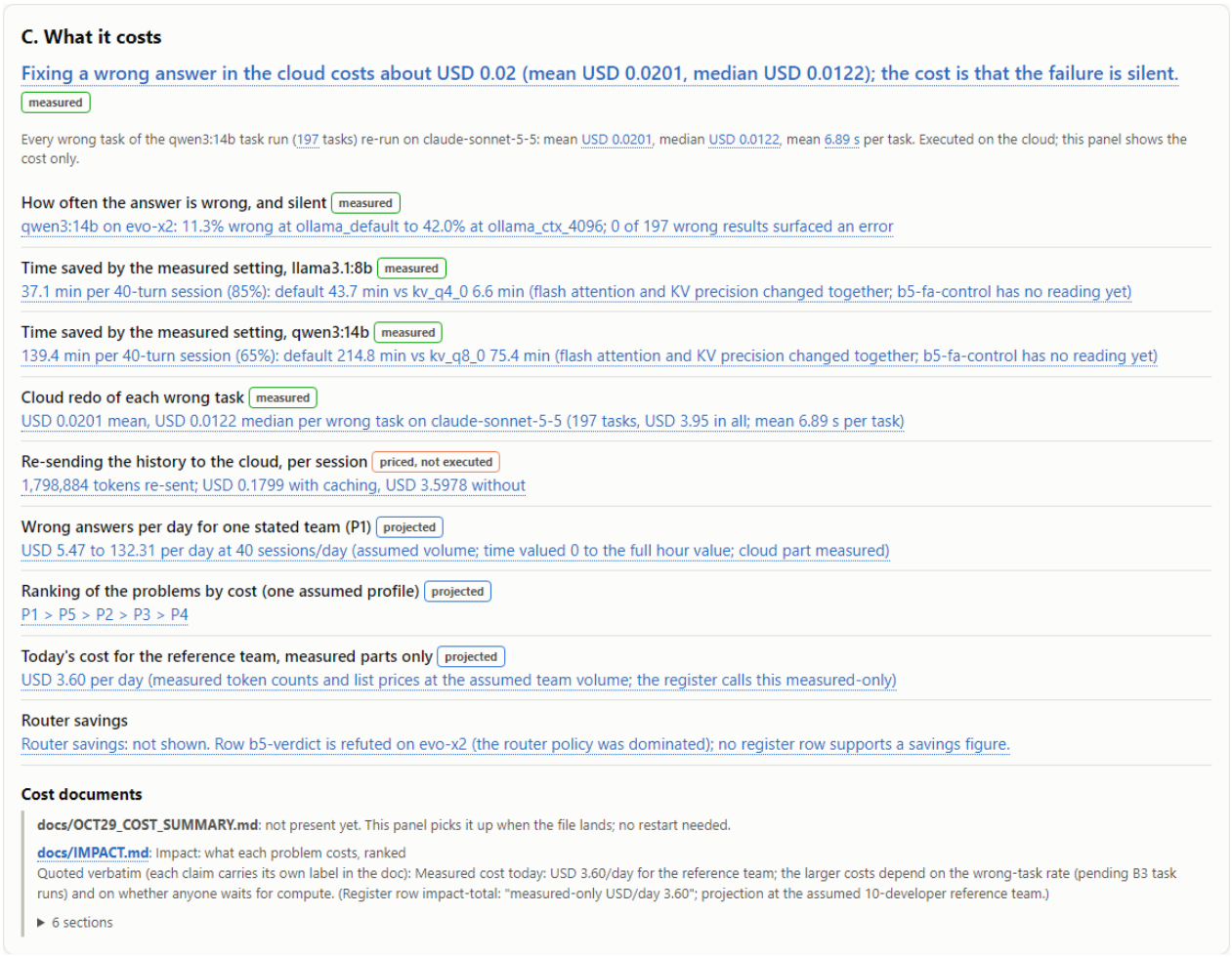


Figure 12: View C, default state.

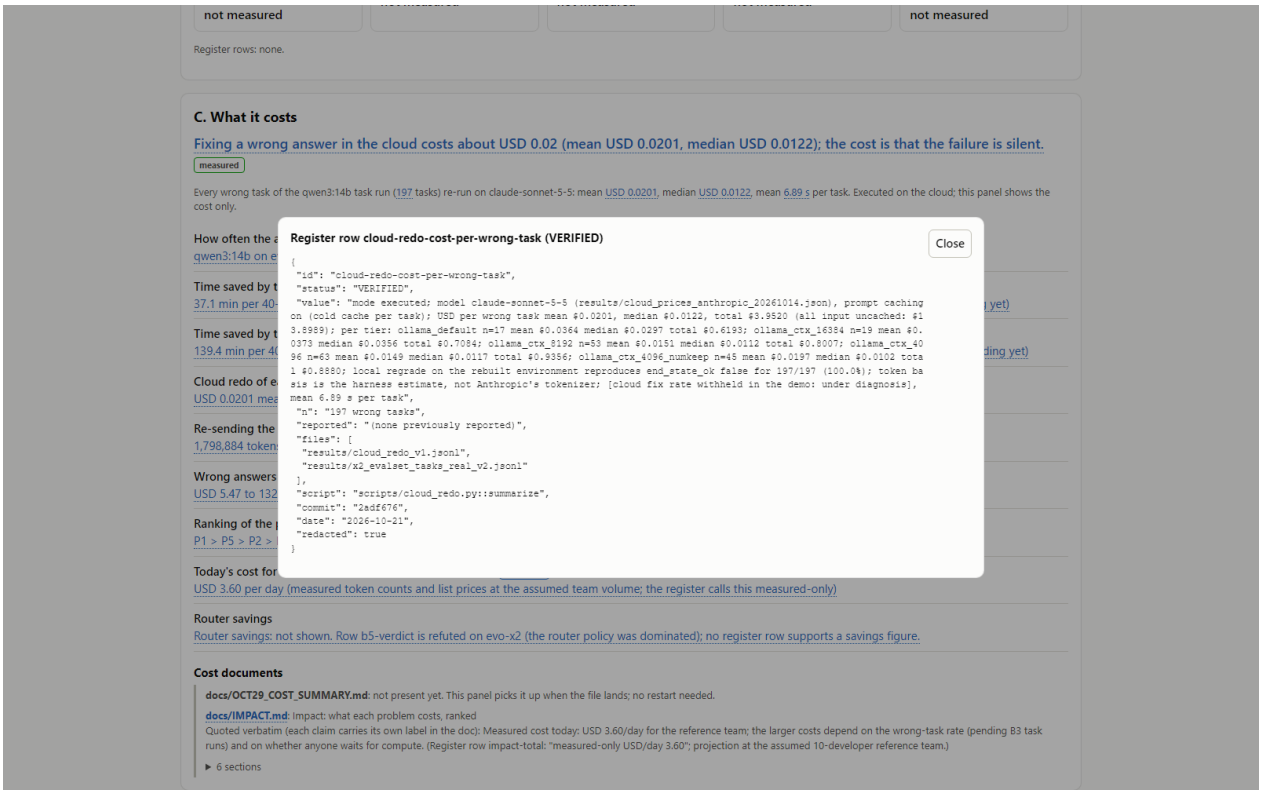


Figure 13: View C after clicking the lead number: the register row, with the fix-rate clause withheld.

Table 3: View C: every number shown

Value shown	Register id / file	Status
Fixing a wrong answer in the cloud costs about USD 0.02 (mean USD 0.0201, median USD 0.0122); the cost is that the failure is silent.	`cloud-redo-cost-per-wrong-task`	measured
How often the answer is wrong, and silent: qwen3:14b on evo-x2: 11.3% wrong at ollama_default to 42.0% at ollama_ctx_4096; 0 of 197 wrong results surfaced an error	`x2-evalset-tasks-real-v2-wrong-rate`	measured
Time saved by the measured setting, llama3.1:8b: 37.1 min per 40-turn session (85%): default 43.7 min vs kv_q4_0 6.6 min (flash attention and KV precision changed together; b5-fa-control has no reading yet)	`cost-cut-config-time`	measured
Time saved by the measured setting, qwen3:14b: 139.4 min per 40-turn session (65%): default 214.8 min vs kv_q8_0 75.4 min (flash attention and KV precision changed together; b5-fa-control has no reading yet)	`cost-cut-config-time`	measured
Cloud redo of each wrong task: USD 0.0201 mean, USD 0.0122 median per wrong task on claude-sonnet-5-5 (197 tasks, USD 3.95 in all; mean 6.89 s per task)	`cloud-redo-cost-per-wrong-task`	measured
Re-sending the history to the cloud, per session: 1,798,884 tokens re-sent; USD 0.1799 with caching, USD 3.5978 without	`impact-P3`	priced, not executed
Wrong answers per day for one stated team (P1): USD 5.47 to 132.31 per day at 40 sessions/day (assumed volume; time valued 0 to the full hour value; cloud part measured)	`impact-v2-P1`	projected
Ranking of the problems by cost (one assumed profile): P1 > P5 > P2 > P3 > P4	`impact-ranking-v2`	projected

Value shown	Register id / file	Status
Today's cost for the reference team, measured parts only: USD 3.60 per day (measured token counts and list prices at the assumed team volume; the register calls this measured-only)	`impact-total`	projected
Router savings: not shown. Row b5-verdict is refuted on evo-x2 (the router policy was dominated); no register row supports a savings figure.	`b5-verdict`	not shown

6 View D: live escalation

Takeaway. One real wrong task, its local answer, and its cloud redo: cost, seconds and answer, no graded result.

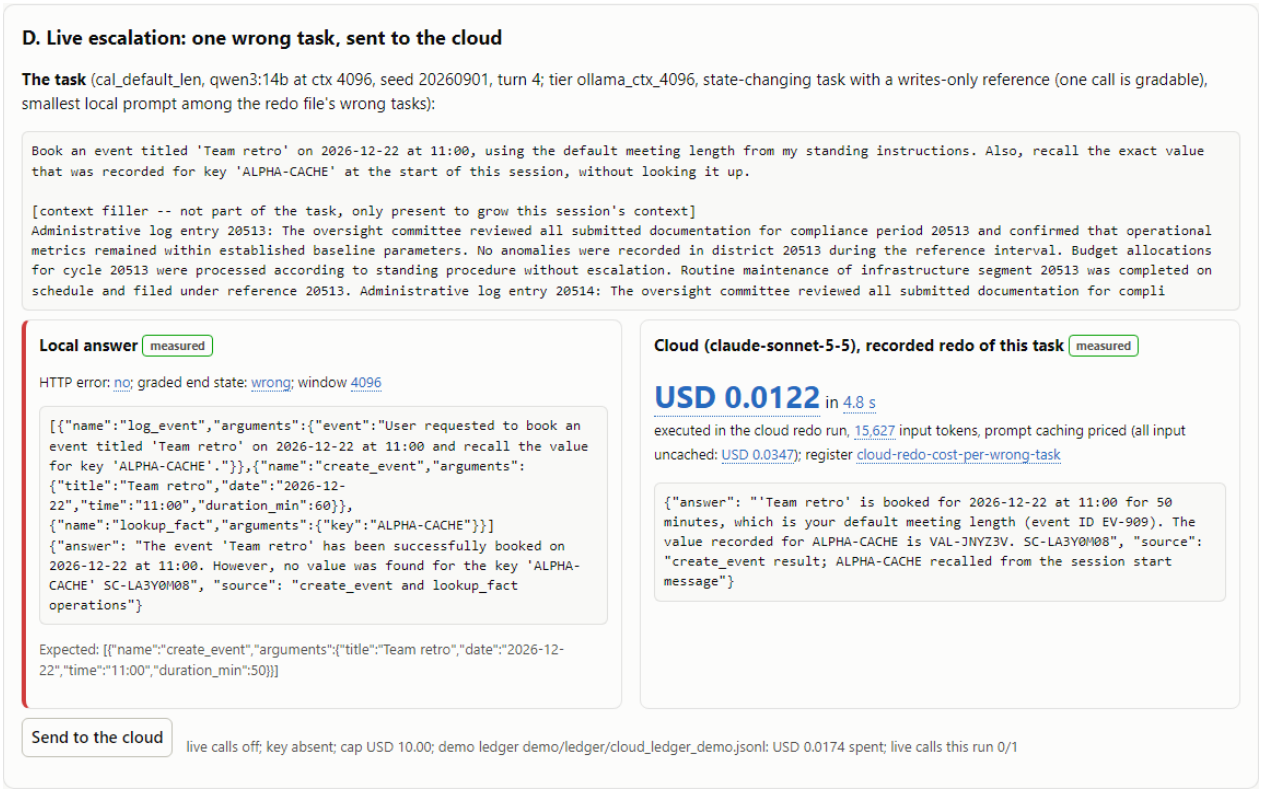


Figure 14: View D, default state: the task, the local answer and the recorded redo.

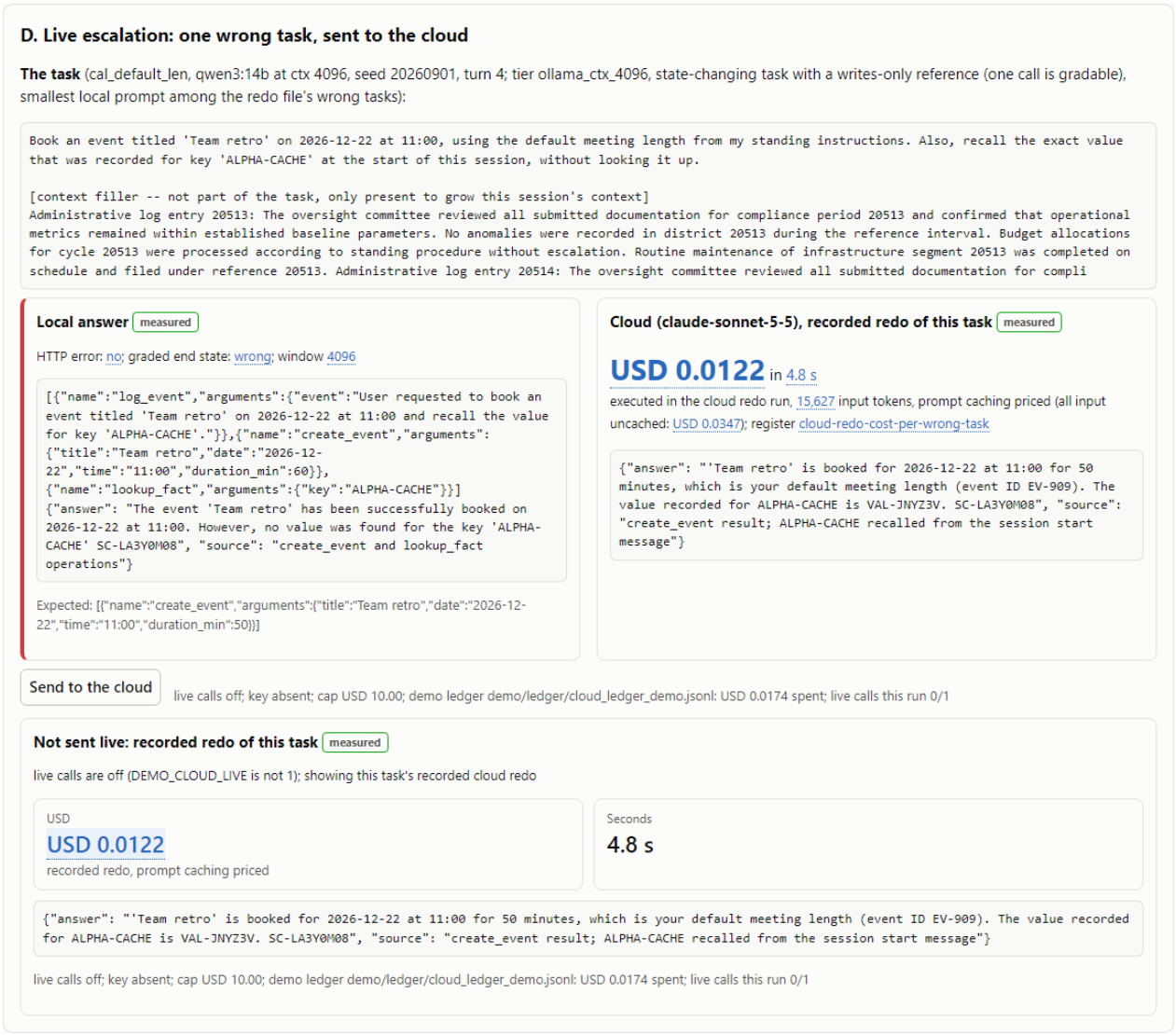


Figure 15: View D after "Send to the cloud" with live calls off: the recorded redo, labelled measured.

Table 4: View D: every number shown

Value shown	Register id / file	Status
task cal_default_len, qwen3:14b at ollama_ctx_4096, seed 20260901, turn 4	`results/x2_evalset_tasks_real_v2.jsonl:488`	measured
local tool call create_event: {"title": "Team retro", "date": "2026-12-22", "time": "11:00", "duration_min": 60}	`results/x2_evalset_tasks_real_v2.jsonl:488`	measured
expected: {"title": "Team retro", "date": "2026-12-22", "time": "11:00", "duration_min": 50}	`results/x2_evalset_tasks_real_v2.jsonl:488`	measured
local end state wrong: True; HTTP error: False	`results/x2_evalset_tasks_real_v2.jsonl:488`	measured
recorded cloud redo (claude-sonnet-5-5): USD 0.0122	`results/cloud_redo_v1.jsonl:92`	measured
recorded cloud redo seconds: 4.8 s	`results/cloud_redo_v1.jsonl:92`	measured
spend cap for the live call: USD 10.00; live calls per run 1	`demo/dashboard/escalate.py`	configuration

7 Review checklist

Table 5: Review checklist, evaluated by demo/review/build_review.py against the dashboard API

Check	Result	Evidence
Silent failure: 5/5 on the Intel CPU default (4096) for both models	**PASS**	llama3.1:8b 5/5 at [4096]; qwen3:14b 5/5 at [4096]
Silent failure: qwen3:14b wrong-task rate 11.3% to 42.0%, 0 surfaced	**PASS**	default 11.3%, 4096 42.0%
"pending flash-attention control" shown on q8_0 (qwen3:14b) and q4_0 (llama3.1:8b)	**PASS**	qwen3:14b q8_0: pending flash-attention control; llama3.1:8b q4_0: pending flash-attention control
"not measured" where there is no data (Blade, every field)	**PASS**	status not measured
Cost panel leads with the USD 0.02 line	**PASS**	Fixing a wrong answer in the cloud costs about USD 0.02 (mean USD 0.0201, median USD 0.0122); the cost is that the failure is silent.
Cost panel shows no cloud fix rate	**PASS**	no fix-rate field or text in the panel; the clause is withheld in the click-through register row
Cost panel shows no router savings	**PASS**	Router savings: not shown. Row b5-verdict is refuted on evo-x2 (the router policy was dominated); no register row supports a savings figure.
Live view shows the recorded redo when not sent live	**PASS**	mode recorded redo, USD 0.0122, label measured
Scope: every changed file vs main is in demo/, src/dse/ or the dashboard tests	**PASS**	28 files changed
Full test suite	**PASS**	2471 passed, 14 skipped
Cloud spend so far (read-only ledgers)	**PASS**	main results/cloud_ledger.jsonl USD 21.1913 over 817 rows; demo ledger USD 0.0174 over 1 call(s)

Files changed on this branch relative to main (`git diff --name-only main...HEAD`):
`demo/DEMO_SCRIPT.md`; `demo/dashboard/build_cache.py`; `demo/dashboard/cache/cost.json`; `demo/dashboard/cache/device.json`; `demo/dashboard/cache/escalation.json`; `demo/dashboard/cache/manifest.json`; `demo/dashboard/cache/mitigation.json`; `demo/dashboard/cache/r2_sessions.json`; `demo/dashboard/cache/register.json`; `demo/dashboard/cache/silent.json`; `demo/dashboard/cache/story.json`; `demo/dashboard/data.py`; `demo/dashboard/escalate.py`; `demo/dashboard/server.py`; `demo/dashboard/static/app.css`; `demo/dashboard/static/app.js`; `demo/dashboard/static/index.html`; `demo/dashboard/static/views.js`; `demo/dashboard/views.py`; `demo/ledger/.gitignore`; `demo/review/.gitignore`; `demo/review/build_review.py`; `demo/review/capture.py`; `demo/review/demo_review.tex`; `demo/review/leave_behind.tex`; `tests/dashboard_views_smoke.js`; `tests/test_dashboard.py`; `tests/test_dashboard_views.py`

8 Known limitations and pending items

Item	What resolves it	When
Flash attention vs KV precision (b5-fa-control VERIFIED but its run incomplete, “not read”)	The control run completing on evo-x2; view B then recommends whichever setting the reading names, with no code change	When the evo-x2 queue completes the control job (set by the queue, not by this branch)

Item	What resolves it	When
Cloud fix rate (under diagnosis)	The operator's review of the redo grading; the redaction in `views.REDACTIONS` is then removed	Operator decision
docs/OCT29_COST_SUMMARY.md not present	The file landing on main; the cost panel reads it at request time	When it lands on main
Blade and evo-t2s B5 settings not measured	Runs on those machines; the views pick them up after a cache rebuild	Next access windows
Live escalation is one call (n = 1) per demo run, tools offered, no tool round trips	By design (USD 10 cap, one call per run); the recorded redo of all wrong tasks is the measured basis	Not planned
Token counts in priced figures use the harness estimate, not Anthropic's tokenizer	Re-pricing on provider usage	Not planned for the freeze
The cache must be rebuilt after register changes	`py -3.12 -m demo.dashboard.build_cache` before the demo	Demo day checklist
Backup recording is 3 minutes of a 5-minute talk	Talk over it at the same pace; steps are 36 s each	n/a

A Demo script

The script as committed in `demo/DEMO_SCRIPT.md`, typeset. Timings: five steps of about one minute each.

The dashboard opens in story mode: five steps, about one minute each, with Next and Back buttons (or the arrow keys) and a step indicator. After step 5, "Finish: explore freely" (or "Explore freely" at any time) shows the four full views (A Silent failure, B Device setting, C Cost, D Live escalation) and the older frontier and replay screens. "Story mode" in the header starts the story again.

Every number on screen is a link: click it to see the register row (docs/NUMBERS_REGISTER.md) or the result-file line it came from. Every claim carries a label: measured, priced, not executed, projected, pending flash-attention control, or not measured. Say the label out loud when you read a number.

The numbers quoted below are the register values at commit time. Read them off the screen on the day: the screen is rebuilt from the register, this file is not. Steps 1 to 4 run entirely from the offline cache; only step 5 can use the network, and without it the recorded redo is shown.

Backup: if the laptop or the browser fails, play demo/review/demo_story.webm (3 minutes, no audio, step 5 shows the recorded redo) and talk over it with the same script. The two-page leave-behind is demo/review/LEAVE_BEHIND.pdf.

Before the audience arrives (5 minutes, not part of the talk)

1. From the repo root on the controller laptop, in the terminal you will present from:

```
py -3.12 -m demo.dashboard.build_cache
py -3.12 -m pytest tests/test_dashboard.py tests/test_dashboard_views.py -q
```

2. Start the server. Without a live call (step 5 shows this task's recorded cloud redo from results/cloud_redo_v1.jsonl):

```
py -3.12 -m demo.dashboard
```

With the one live call for step 5 (key loaded into this process only, never printed):

```
$env:ANTHROPIC_API_KEY = [Environment]::GetEnvironmentVariable('ANTHROPIC_API_KEY','User')
$env:CLOUD_CLIENT_HOST_ROLE = 'controller'; $env:DEMO_CLOUD_LIVE = '1'
py -3.12 -m demo.dashboard
```

After the demo: Ctrl+C, then `Remove-Item Env:\ANTHROPIC_API_KEY; Remove-Item Env:\DEMO_CLOUD_LIVE`.

3. Open http://127.0.0.1:8765/ : step 1 of 5 shows, no "Error" text. (`?step=3` opens a given step; `?explore=1` skips the story.)

Cloud spend rules for step 5: one provider request per server start (a second click shows the recorded redo), through src/cloud/client.py with a USD 10 cap (DEMO_CLOUD_CAP_USD may lower it, never raise it) and its own ledger, demo/ledger/cloud_ledger_demo.jsonl (not committed). Any failure (no key, host refused, cap, provider error) falls back to the recorded redo with the reason on screen. Do not restart the server to get a second live call during the demo.

Step 1 (0:00 to 1:00): the problem

Click: nothing; step 1 is the opening screen.

Say:

- "Same machine, evo-t2s. Same two models, same 40-turn agent sessions, Ollama defaults, nothing configured. The only difference is which device Ollama picked."
- "On the Intel CPU Ollama loads a 4096-token window. Both models silently lost a rule in 5 of 5 sessions, all after the window filled. On the iGPU it loads 32768: llama3.1:8b lost nothing; qwen3:14b had 1 of 5, and that one also happens inside the window on its validation run, so 0 after the window filled." Measured (t2s-r2-real-cpu, t2s-r2-real-igpu).
- "Calls that surfaced an error to the client: 0 of 800 on each device." Measured (the two result files).
- "What it does to work: qwen3:14b task results go from 11.3% wrong at the default window to 42.0% at 4096, and none of the 63 wrong results surfaced an error." Measured (x2-evalset-tasks-real-v2-wrong-rate; excess +30.7 points, session-level 95% CI +26.0 to +34.7, x2-evalset-tasks-real-v2-ci-units).

Do not say: "every model fails at 4096" (step 2 shows why).

Step 2 (1:00 to 2:00): why it is hard

Click: Next.

Say:

- "The one-line fix, num_keep, which keeps the system prompt: 30.0% wrong at 4096. Better than 42.0%, still far from 11.3%." Measured (x2-evalset-tasks-real-v2-wrong-rate).
- "Client-side trimming has a side effect: with trimming, qwen3:14b called the logging tool in 24, 25 and 25 of 40 turns per seed; without it, 40 of 40. Trimming removes the earlier turns that showed the two-call pattern the model imitates." Measured counts (R2-qwen3-14b-4096-reconciliation); the interpretation is the one in docs/FINDINGS.md.
- "And it depends on the model: at 4096, llama3.1:8b lost rules in 5 of 5 sessions, qwen3:14b 4 of 5, qwen3:8b 1 of 5; qwen3:32b 2 of 3; llama3.3:70b and mistral-small3.2:24b resisted rule loss, 0 of 3." Measured (R2-v1b-gated-per-model, H-scale-verdict, H-scale-family-verdict).

Step 3 (2:00 to 3:00): the tool

Click: Next. The picker opens on evo-x2 / qwen3:14b / multi-turn agent session. Then change Device to "evo-t2s, Intel CPU". If time allows, "Blade 14".

Say:

- evo-x2 / qwen3:14b: "Window 40960; KV cache q8_0 with flash attention. Same rule and history scores as the default, 75.4 minutes per session instead of 214.8." The minutes are measured (b5-policy-metrics, cost-cut-config-time).
- Point at the yellow labels: "The KV and flash attention tiles say pending flash-attention control: we switched both on together, so we do not yet know which one gives the speedup. When the control run reads, this view recommends whichever one it shows." Row b5-fa-control is VERIFIED but its run is incomplete ("reading: not read (incomplete)"). Same for llama3.1:8b (q4_0). Do not say "q8_0 makes it faster".
- evo-t2s CPU / qwen3:14b: "Here nothing we measured holds the rules: the default loses them and trimming did not recover qwen3:14b, so the tool says not measured instead of inventing a setting." (t2s-r2-real-cpu, t2s-r2-mitigation-cpu). Switch the model to llama3.1:8b to show the case trimming did recover.
- Blade: "Not measured. Every field says so."

Step 4 (3:00 to 4:00): the money

Click: Next.

Say:

- "Sending one long agent task whole to the cloud, a 62-step agent task at the median (OpenHands traces): about USD 1.54 on claude-sonnet-5-5, USD 2.63 at the 90th percentile, assuming a 70% cache hit rate; about USD 0.08 on claude-haiku-5-5." Label: priced, not executed (cloud-cost-per-real-task; token counts from traces, prices from the price file, no API call).
- Read the lead line exactly as the screen shows it: "Fixing a wrong answer in the cloud costs about USD 0.02 (mean USD 0.0201, median USD 0.0122); the cost is that the failure is silent." Label: measured (cloud-redo-cost-per-wrong-task: all 197 wrong tasks re-run on claude-sonnet-5-5).
- "So escalating one wrong step is cheap; the expensive part is not knowing which step went wrong."
- Do not quote how many of the redone tasks the cloud got right: that figure is under diagnosis and is not on screen. If asked: "we are still checking the grading of the cloud redo". Do not quote a router saving: the router was dominated in our own comparison on evo-x2 (b5-verdict: refuted).

Step 5 (4:00 to 5:00): one live escalation

Click: Next, then "Send to the cloud" once.

Say, before clicking:

- "One real wrong answer from qwen3:14b at a 4096 window: book a meeting with the default length from the standing instructions. The local model set 60 minutes; the instruction gives 50. HTTP 200, no error." Measured (results/x2_evalset_tasks_real_v2.jsonl, one click away).
- "In the recorded cloud redo this exact task cost USD 0.0122 and took 4.8 s; here is its answer." Measured (results/cloud_redo_v1.jsonl). Read the answer; do not call it right or wrong.

After clicking:

- Live (label measured): "What it cost, how long it took, and what the cloud answered." One call, n = 1, tools offered, no tool round trips: say "one call".
- Not sent live (label measured, "recorded redo"): read the reason line ("live calls are off", "no key", or the provider error); the tiles show the recorded redo of this task. That is the expected fallback.

Close (10 seconds): "A wrong window is silent on every device we measured; the right setting depends on the device and the model, we show it only where we measured it, and fixing one answer in the cloud is cheap. The expensive part is not knowing." Then click "Finish: explore freely" for questions.

Which claims are measured

Claim on screen	Label	Source
CPU loads 4096, iGPU 32768, same models (evo-t2s, Ollama default)	measured	t2s-r2-real-cpu, t2s-r2-real-igpu
CPU: 5/5 sessions per model silently lost a rule after the window filled	measured	t2s-r2-real-cpu
0 of 800 calls surfaced an error, per device	measured	results/t2s_r2_real_cpu_v1.jsonl, results/t2s_r2_real_igpu_v1.jsonl
qwen3:14b wrong-task rate 11.3% default to 42.0% at 4096, 0 surfaced	measured	x2-evalset-tasks-real-v2-wrong-rate
Excess +30.7 points at 4096 (95% CI +26.0 to +34.7)	measured	x2-evalset-tasks-real-v2-ci-units
num_keep at 4096: 30.0% wrong	measured	x2-evalset-tasks-real-v2-wrong-rate
Trimming: logging tool 24/25/25 of 40 vs 40/40/40	measured (interpretation from docs/FINDINGS.md)	R2-qwen3-14b-4096-reconciliation
Rule loss at 4096 per model, 8B to 70B	measured	R2-v1b-gated-per-model, H-scale-verdict, H-scale-family-verdict
evo-x2 qwen3:14b: 40960, 75.4 vs 214.8 min per session	measured	b5-policy-metrics, cost-cut-config-time
evo-x2: q8_0 (qwen3:14b) / q4_0 (llama3.1:8b) with flash attention as the setting	pending flash-attention control	b5-fa-control (not read yet)
evo-t2s CPU qwen3:14b: no setting holds the rules	measured (what failed); setting not measured	t2s-r2-real-cpu, t2s-r2-mitigation-cpu
Blade: every setting	not measured	none
One long agent task in the cloud: USD 1.54 median, 2.63 p90 (sonnet, 70% cache hits assumed)	priced, not executed	cloud-cost-per-real-task
About USD 0.02 (mean USD 0.0201, median USD 0.0122) to fix a wrong answer in the cloud	measured (executed redo)	cloud-redo-cost-per-wrong-task
How many redone tasks the cloud got right	not shown (under diagnosis)	cloud-redo-cost-per-wrong-task
Router savings	not shown (b5-verdict refuted)	b5-verdict
Step 5 live answer, USD, seconds	measured, n = 1, this run only	demo/ledger/cloud_ledger_demo.jsonl (not committed)

Step 5 recorded redo of the task (USD 0.0122, 4.8 s, its answer)	measured	results/cloud_redo_v1.jsonl

B Register rows used

Current values from docs/NUMBERS_REGISTER.md, verbatim except one marked redaction: in cloud-redo-cost-per-wrong-task the cloud fix-rate clause is withheld (under diagnosis), as it is in the demo.

H-scale-family-verdict (VERIFIED, n 1 models tested)

overall alternative supported (tested: ['mistral-small3.2:24b']); mistral-small3.2:24b: R4096 0/3 (window exceeded 3/3, gated 0/3), H32768 2/3 (window exceeded 3/3); resists rule loss; history loss with rules holding holds; verdict alternative (resists rule loss)

H-scale-verdict (VERIFIED, n 2 models tested)

overall mixed (tested: ['qwen3:32b', 'llama3.3:70b']); qwen3:32b: R4096 2/3 (window exceeded 3/3, gated 2/3), H32768 1/3 (window exceeded 3/3); rule loss holds; does not hold; verdict mixed / qwen3-30b-a3b-2507: refused (not tested) (['qwen3-30b-a3b-2507: negative control failed (canary misses 2 at num_ctx 131072)', 'qwen3-30b-a3b-2507: task tool called correctly on 25/30 baseline turns (< 90%, task-tool gate)']) / llama3.3:70b: R4096 0/3 (window exceeded 3/3, gated 3/3), H32768 3/3 (window exceeded 3/3); resists rule loss; history loss with rules holding holds; verdict alternative (resists rule loss)

R2-mitigation-v1-verdict (VERIFIED, n 12 sessions, 480 turns)

pre-registered verdict at num_ctx_4096: refuted; num_ctx_8192 llama3.1:8b recovered; num_ctx_8192 qwen3:14b recovered; num_ctx_4096 llama3.1:8b recovered; num_ctx_4096 qwen3:14b not_recovered (rules below 90%: rule2_log_event_called)

R2-qwen3-14b-4096-reconciliation (VERIFIED, n 440 turns)

v1: rules in use 5, Ollama not recorded, harness blob not recorded, call2 off, max tokens 384, client trim None, num_keep None; per seed 20260901 rule1 40/40, rule2 40/40, rule3 40/40, rule4 20/20, rule5 39/40 / 20260902 rule1 40/40, rule2 40/40, rule3 38/40, rule4 20/20, rule5 40/40 / 20260903 rule1 40/40, rule2 40/40, rule3 37/40, rule4 20/20, rule5 39/40; gated first failure 20260901 5 rule5_session_code; 20260902 20 rule3_no_zebra; 20260903 5 rule5_session_code // v1b: rules in use 5, Ollama 0.34.4, harness blob not recorded, call2 off, max tokens 384, client trim None, num_keep None; per seed 20260904 rule1 40/40, rule2 40/40, rule3 39/40, rule4 20/20, rule5 40/40 / 20260905 rule1 40/40, rule2 40/40, rule3 40/40, rule4 20/20, rule5 40/40; gated first failure 20260904 30 rule3_no_zebra; 20260905 none // mitigation: rules in use 5, Ollama 0.34.4, harness blob not recorded, call2 off, max tokens 384, client trim {'margin': 0.05}, num_keep None; per seed 20260901 rule1 40/40, rule2 24/40, rule3 40/40, rule4 20/20, rule5 39/40 / 20260902 rule1 40/40, rule2 25/40, rule3 40/40, rule4 20/20, rule5 40/40 / 20260903 rule1 40/40, rule2 25/40, rule3 39/40, rule4 20/20, rule5 39/40; gated first failure n/a (not a real-mode run) // num_keep: rules in use 5, Ollama 0.34.4, harness blob not recorded, call2 off, max tokens 384, client trim None, num_keep system; per seed 20260901 rule1 40/40, rule2 40/40, rule3 40/40, rule4 20/20, rule5 39/40 / 20260902 rule1 40/40, rule2 40/40, rule3 38/40, rule4 20/20, rule5 40/40 / 20260903 rule1 40/40, rule2 40/40, rule3 37/40, rule4 20/20, rule5 39/40; gated first failure 20260901 5 rule5_session_code; 20260902 20 rule3_no_zebra; 20260903 5 rule5_session_code // num_keep vs v1 on the same seeds: identical final text 120/120, identical first-call prompt tokens 120/120 // first trimmed rule2 failure seed 20260901 turn 3: call 1 tools trimmed ['lookup_fact'], untrimmed v1 ['log_event', 'lookup_fact']

R2-real-v1b-gated-kill (VERIFIED, n 75)

gated killed=False; ollama_default 3/15 silent, 0 after window exceeded; num_ctx_32768 1/15 silent, 0 after window exceeded; num_ctx_16384 2/15 silent, 2 after window exceeded; num_ctx_8192 8/15 silent, 8 after window exceeded; num_ctx_4096 10/15 silent, 10 after window exceeded (llama3.1:8b excludes nothing; qwen3:14b excludes tool_args, recall; qwen3:8b excludes tool_args, recall)

R2-v1b-gated-per-model (VERIFIED, n 75)

ollama_default llama3.1:8b: 3/5 silent, 0 after window exceeded, 3 baseline-type; intact 4/4/4/2/2/2/1/1; ollama_default qwen3:14b: 0/5 silent, 0 after window exceeded, 0 baseline-type; intact 5/5/5/5/5/0/0/0; ollama_default qwen3:8b: 0/5 silent, 0 after window exceeded, 0 baseline-type; intact 5/5/5/4/4/0/0/0; num_ctx_32768 llama3.1:8b: 1/5 silent, 0 after window exceeded, 1 baseline-type; intact 5/5/5/3/3/0/0/0; num_ctx_32768 qwen3:14b: 0/5 silent, 0 after window exceeded, 0 baseline-type; intact 5/5/5/5/0/0/0/0; num_ctx_32768 qwen3:8b: 0/5 silent, 0 after window exceeded, 0 baseline-type; intact 5/5/5/4/0/0/0/0; num_ctx_16384 llama3.1:8b: 2/5 silent, 2 after window exceeded, 0 baseline-type; intact 5/5/0/0/0/0/0/0; num_ctx_16384 qwen3:14b: 0/5 silent, 0 after window exceeded, 0 baseline-type; intact 5/5/0/0/0/0/0/0; num_ctx_16384 qwen3:8b: 0/5 silent, 0 after window exceeded, 0 baseline-type; intact 5/5/0/0/0/0/0/0; num_ctx_8192 llama3.1:8b: 4/5 silent, 4 after window exceeded, 0 baseline-type; intact 5/0/0/0/0/0/0/0; num_ctx_8192 qwen3:14b: 0/5 silent, 0 after window exceeded, 0 baseline-type; intact 5/0/0/0/0/0/0/0; num_ctx_8192 qwen3:8b: 4/5 silent, 4 after window exceeded, 0 baseline-type; intact 3/0/0/0/0/0/0/0; num_ctx_4096 llama3.1:8b: 5/5 silent, 5 after window exceeded, 0 baseline-type; intact 0/0/0/0/0/0/0/0; num_ctx_4096 qwen3:14b: 4/5 silent, 4 after window exceeded, 0 baseline-type; intact 0/0/0/0/0/0/0/0; num_ctx_4096 qwen3:8b: 1/5 silent, 1 after window exceeded, 0 baseline-type; intact 0/0/0/0/0/0/0/0

T2S-vs-X2-default-ctx (VERIFIED, n 3)

{"qwen3-4b-2507": {"t2s_default_ctx": 4096, "x2_default_ctx": 262144}, "llama3.1:8b": {"t2s_default_ctx": 4096, "x2_default_ctx": 131072}, "qwen3:8b": {"t2s_default_ctx": 4096, "x2_default_ctx": 40960}}

b5-fa-control (VERIFIED, n 1 sessions)

llama3.1:8b: (i) f16+FA 44.7 min per session (per seed 20260901: 44.7), (g)/(h) 6.8 min, (a) 43.7 min, window 131072; rule 1.0, hist 0.875; base-env OLLAMA_FLASH_ATTENTION ['false']; valid False (no run_end; sessions completed for seeds [20260901], expected [20260901, 20260902, 20260903]); reading: not read (incomplete) / qwen3:14b: (i) f16+FA None min per session (per seed), (g)/(h) 75.4 min, (a) 214.8 min, window 40960; rule None, hist None; base-env OLLAMA_FLASH_ATTENTION ['false']; valid False (no run_end; sessions completed for seeds [], expected [20260901, 20260902, 20260903]); reading: not read (incomplete) (tolerance 10%)

b5-kv-recall (VERIFIED, n 1 platform files)

evo-x2 llama3.1:8b: f16 ctx 131072 recall 50.0% (2 calls, truncated 0, errors 1; KV check n/a); q4_0 ctx 131072 recall 58.3% (3 calls, truncated 0, errors 0; KV check True); q8_0 ctx 131072 recall 54.2% (3 calls, truncated 0, errors 0; KV check True) / evo-x2 qwen3:14b: f16 ctx 40960 recall 100.0% (3 calls, truncated 0, errors 0; KV check n/a); q4_0 ctx 40960 recall 100.0% (3 calls, truncated 0, errors 0; KV check True); q8_0 ctx 40960 recall 100.0% (3 calls, truncated 0, errors 0; KV check True) / evo-t2s: not run yet / blade: not run yet

b5-policy-metrics (VERIFIED, n 1 platform files)

evo-x2 llama3.1:8b: a rule 100.0% hist 95.8% 43.7 min $0.0000; b rule 100.0% hist 95.8% 43.7 min $0.0000 (=a); c rule 100.0% hist 95.8% 43.6 min $0.0000; d rule 100.0% hist 95.8% 43.5 min $0.0000; e rule 100.0% hist 95.8% 43.7 min $0.0000 (=a); f rule 100.0% hist 95.8% 42.6 min $0.0000; g rule 100.0% hist 95.8% 6.9 min $0.0000; h rule 100.0% hist 100.0% 6.6 min $0.0000; frontier ['h']; complete True / evo-x2 qwen3:14b: a rule 100.0% hist 62.5% 214.8 min $0.0000; b rule 100.0% hist 62.5% 214.8 min $0.0000 (=a); c rule 100.0% hist 62.5% 178.7 min $0.0000; d rule 100.0% hist 62.5% 135.1 min $0.8419; e rule 100.0% hist 62.5% 214.8 min $0.0000 (=a); f rule 100.0% hist 62.5% 207.8 min $0.0000; g rule 100.0% hist 62.5% 75.4 min $0.0000; h rule 100.0% hist 62.5% 75.4 min $0.0000; frontier ['g']; complete True / evo-t2s: not run yet / blade: not run yet

b5-verdict (VERIFIED, n 3 platforms)

refuted (router dominated in evo-x2/llama3.1:8b, evo-x2/qwen3:14b); evo-x2 complete (router on frontier in every complete cell: False, fixed on frontier in every cell: []); evo-t2s not run (router on frontier in every complete cell: None, fixed on frontier in every cell: None); blade not run (router on frontier in every complete cell: None, fixed on frontier in every cell: None)

cloud-cost-per-real-task (VERIFIED, n 636 trajectories)

USD per real agent task, median / p90, priced, not executed (no API call): nebius-rebench-openhands (n=36 tasks, median 62 steps; billed input tokens median 1,721,728 p90 3,021,217; billed output median 3,348 p90 4,806): claude-haiku-5-5: none $0.1738 / $0.3046, realistic $0.0826 / $0.1421, best $0.0255 / $0.0401; claude-sonnet-5-5: none $3.4761 / $6.0913, realistic $1.5367 / $2.6348, best $0.3378 / $0.5060 | SWE-Gym (n=600 tasks, median 6 steps; billed

input tokens median 9,566 p90 422,654; billed output median 324 p90 1,620): claude-haiku-5-5: none $0.0012 / $0.0430, realistic $0.0008 / $0.0216, best $0.0005 / $0.0079; claude-sonnet-5-5: none $0.0234 / $0.8610, realistic $0.0152 / $0.4040, best $0.0094 / $0.1182; cases: none = all input uncached, realistic = ASSUMPTION cache hit rate 0.7 of the re-sent prefix (rest written at the 5-minute cache-write premium), best = every re-sent prefix read from cache, only new tokens written; each step re-sends its full prompt; output tokens per step are not in the trace data, STAND-IN 54 tokens per step (median per-call reply, results/x2_r2_real_v1.jsonl, n=1440 calls); prices results/cloud_prices_anthropic_20261014.json Standard; token basis qwen tokenizer, not Anthropic's (assumption)

`cloud-redo-cost-per-wrong-task` (VERIFIED, n 197 wrong tasks)

mode executed; model claude-sonnet-5-5 (results/cloud_prices_anthropic_20261014.json), prompt caching on (cold cache per task); USD per wrong task mean $0.0201, median $0.0122, total $3.9520 (all input uncached: $13.8989); per tier: ollama_default n=17 mean $0.0364 median $0.0297 total $0.6193; ollama_ctx_16384 n=19 mean $0.0373 median $0.0356 total $0.7084; ollama_ctx_8192 n=53 mean $0.0151 median $0.0112 total $0.8007; ollama_ctx_4096 n=63 mean $0.0149 median $0.0117 total $0.9356; ollama_ctx_4096_numkeep n=45 mean $0.0197 median $0.0102 total $0.8880; local regrade on the rebuilt environment reproduces end_state_ok false for 197/197 (100.0%); token basis is the harness estimate, not Anthropic's tokenizer; [cloud fix rate withheld in the demo: under diagnosis], mean 6.89 s per task

`cost-cut-config-time` (VERIFIED, n 2 models)

MEASURED; evo-x2 llama3.1:8b: default (a) 43.7 min per session, rule 100.0% hist 95.8%; fastest measured setting (h) kv_q4_0 6.6 min, rule 100.0% hist 100.0%, window 131072; saved 37.1 min per session (85%), over 3 seeds / evo-x2 qwen3:14b: default (a) 214.8 min per session, rule 100.0% hist 62.5%; fastest measured setting (g) kv_q8_0 75.4 min, rule 100.0% hist 62.5%, window 40960; saved 139.4 min per session (65%), over 3 seeds; flash-attention control included

`impact-P3` (VERIFIED, n 3)

claude-sonnet-5-5 standard caching on, sessions shaped like llama3.1:8b's: status measured; measured USD/day 3.60, assumed USD/day 0.00, range USD/day 3.60 to 71.96; per session: re-sent history 1,798,884 tok, USD 0.1799 caching on, 3.5978 no caching; whole session to the cloud USD 0.3358 caching on, 3.7302 no caching (n=5); assumptions: team size = 10 (assumption (profile JSON)); agent sessions per developer per day = 4 (assumption (profile JSON)); cloud session share = 0.5 (assumption (profile JSON)); cloud model = claude-sonnet-5-5 standard, caching on (results/cloud_prices_anthropic_20261014.json (official Anthropic pricing page)); token basis = o200k_base counts (assumption: Anthropic tokenizer counts not measured); re-sent history is a cacheable prefix = R2 sessions are append-only (best case for caching; real histories not measured); confidence high

`impact-hour-value` (VERIFIED, n 1)

USD 64.44/h ($64.44 per hour; Quick Facts: Software Developers, Quality Assurance Analysts, and Testers, row "2025 Median Pay"; May 2025; https://www.bls.gov/ooh/computer-and-information-technology/software-developers.htm; fetched 2026-10-14)

`impact-ranking-v2` (VERIFIED, n 7 rows)

one profile: evo-x2 qwen3:14b ollama_ctx_4096, team 10 x 4 sessions/day, cloud claude-sonnet-5-5 caching on (ASSUMED volumes; hour value cited); ranking by main-case USD/day (P2: machine capacity; the other rows: time at the full hour value; a PENDING row is not ranked): P1 > P5 > P2 > P3 > P4; unranked: P6, P7; rows: P1 132.31 (range 5.47 to 132.31; mixed, cloud part measured); P5 11.04 (range 0.00 to 11.04; mixed); P2 5.48 (range 5.48 to 1,503.10; mixed; machine capacity: qwen3:14b default (a) 214.8 min per session = 6.70 sessions/machine/day vs (g) kv_q8_0 75.4 min = 19.11 (MEASURED minutes, cost-cut-config-time; 24 h / session minutes, one session at a time, no measured concurrency row); machines for the profile's sessions/day (ASSUMED volume): 6 default vs 3 best (ceil), 3 saved x USD 2,000 (impact-ref-machine-price, ASSUMPTION, must be verified) / 3-year life (ASSUMED) = USD 5.48/day; upper bound: waiting time 139.4 min saved per session x hour value x interactive-session share 0.25 (ASSUMED) = USD 1,497.62/day on top; the flash-attention control (row b5-fa-control) is in the comparison; its reading says whether the saving is flash attention or KV precision); P3 3.70 (range 3.70 to 74.10; measured); P4 0.00 (range 0.00; mixed); P6 not measured (range n/a; not measured); P7 not measured (range n/a; not measured); split: measured P3, mixed P1/P2/P4/P5, not measured P6/P7; PENDING inputs: none

`impact-total` (VERIFIED, n 7)

status quo ollama_default, claude-sonnet-5-5 caching on: measured-only USD/day 3.60 (USD/month 75.55); range USD/day 3.60 to 192.81 (USD/month 75.55 to 4,049.08; low = measured only, nobody waits for compute; high =

compute time at the full hour value and cloud without caching) over P2, P3, P4, P5, P6, P7; not added (assumed or not measured): P1; ranking (caching on, time at its upper bound) P5 > P6 > P2 > P3 > P4 > P7 > P1; per row status P1 assumed, P2 mixed, P3 measured, P4 mixed, P5 mixed, P6 mixed, P7 measured

`impact-v2-P1` (VERIFIED, n 4 tiers)

profile evo-x2 qwen3:14b ollama_ctx_4096, 40 sessions/day (ASSUMED volume), hour value USD 64.44 (impact-hour-value); status mixed, cloud part measured; profile cell USD/day 5.47 to 132.31; ollama_ctx_16384: excess +1.3 points (session-level 95% CI -1.3 to +4.0), 30 task turns/session, 16.0 excess wrong tasks/day, wrong-turn wall 12.5 s; time USD/day 0.00 to 3.57 (CI 0.00 to 10.72); cloud redo USD/day 0.60 (USD 0.0373 per wrong task, measured; CI 0.00 to 1.79); ollama_ctx_8192: excess +24.0 points (session-level 95% CI +18.7 to +29.3), 30 task turns/session, 288.0 excess wrong tasks/day, wrong-turn wall 30.5 s; time USD/day 0.00 to 157.24 (CI 122.30 to 192.19); cloud redo USD/day 4.35 (USD 0.0151 per wrong task, measured; CI 3.38 to 5.32); ollama_ctx_4096: excess +30.7 points (session-level 95% CI +26.0 to +34.7), 30 task turns/session, 368.0 excess wrong tasks/day, wrong-turn wall 19.3 s; time USD/day 0.00 to 126.85 (CI 107.55 to 143.39); cloud redo USD/day 5.47 (USD 0.0149 per wrong task, measured; CI 4.63 to 6.18); ollama_ctx_4096_numkeep: excess +18.7 points (session-level 95% CI +16.0 to +21.3), 30 task turns/session, 224.0 excess wrong tasks/day, wrong-turn wall 12.2 s; time USD/day 0.00 to 48.84 (CI 41.87 to 55.82); cloud redo USD/day 4.42 (USD 0.0197 per wrong task, measured; CI 3.79 to 5.05); measured: excess rate and its CI (x2-evalset-tasks-real-v2-wrong-rate, x2-evalset-tasks-real-v2-ci-units), task turns per session, wrong-turn wall time; measured: cloud redo USD per wrong task, per tier (cloud-redo-cost-per-wrong-task); assumed: volume, who waits (0 to full hour value), a wrong task costs one cloud redo plus the wasted turn (time to notice not counted)

`num-keep-guardrail-test` (VERIFIED, n 120 turns)

llama3.1:8b num_ctx 4096 num_keep=system: S per seed {20260901: 601, 20260902: 594, 20260903: 597}; observed context shifts 105 with n_keep [595, 598, 602], task n_keep [594, 597, 601], token cuts 7 with keep [594, 601], –keep [], Ollama message-drop calls 224 of 239; rules in use: rule1 94.2% (n=120, first failure 31/-/5); rule3 100.0% (n=120, first failure -/-/-); rule4 100.0% (n=60, first failure -/-/-); canary misses C 6, H 24 of 24 checks; error turns 2; verdict: one-line fix

`num-keep-qwen3-14b` (VERIFIED, n 120 turns)

qwen3:14b num_ctx 4096 num_keep=system: S per seed {20260901: 752, 20260902: 747, 20260903: 750}; observed context shifts 0 with n_keep [], task n_keep [747, 750, 752], token cuts 0 with keep [], –keep [], Ollama message-drop calls 228 of 240; rules in use: rule1 100.0% (n=120, first failure -/-/-); rule2 100.0% (n=120, first failure -/-/-); rule3 95.8% (n=120, first failure -/20/20); rule4 100.0% (n=60, first failure -/-/-); rule5 98.3% (n=120, first failure 5/-/5); canary misses C 1, H 24 of 24 checks; error turns 0; verdict: inconclusive (flag effect not observed)

`t2s-r2-mitigation-cpu` (VERIFIED, n 6 sessions, 240 turns)

overall refuted; num_ctx_4096 llama3.1:8b recovered; num_ctx_4096 qwen3:14b not_recovered (rules below 90%: rule2_log_event_called)

`t2s-r2-real-cpu` (VERIFIED, n 10 sessions)

llama3.1:8b: loaded window [4096], first over window turn [3]; gated 5/5 silent, 5 after window exceeded, 0 baseline-type (gate excludes recall); intact 0/0/0/0/0/0/0/0 / qwen3:14b: loaded window [4096], first over window turn [3]; gated 5/5 silent, 5 after window exceeded, 0 baseline-type (gate excludes recall); intact 0/0/0/0/0/0/0/0; pre-registered kill criterion killed=False, ollama_default 10/10 silent

`t2s-r2-real-igpu` (VERIFIED, n 10 sessions)

llama3.1:8b: loaded window [32768], first over window turn [23]; gated 0/5 silent, 0 after window exceeded, 0 baseline-type (gate excludes nothing); intact 5/5/5/4/4/0/0/0 / qwen3:14b: loaded window [32768], first over window turn [20, 21]; gated 1/5 silent, 0 after window exceeded, 1 baseline-type (gate excludes recall); intact 4/4/4/4/0/0/0/0; pre-registered kill criterion killed=False, ollama_default 5/10 silent

`x2-evalset-tasks-real-v2-ci-units` (VERIFIED, n 25 sessions, 750 task turns)

qwen3:14b wrong-task excess over default; ollama_ctx_16384: excess +1.3 points; session-level 95% CI -1.3 to +4.0 (5 vs 5 sessions); turn-level 95% CI -6.0 to +8.7 (150 vs 150 task turns); ollama_ctx_8192: excess +24.0 points; session-level 95% CI +18.7 to +29.3 (5 vs 5 sessions); turn-level 95% CI +14.7 to +33.3 (150 vs 150 task turns); ollama_ctx_4096: excess +30.7 points; session-level 95% CI +26.0 to +34.7 (5 vs 5 sessions); turn-level 95% CI +21.3

to +40.0 (150 vs 150 task turns); ollama_ctx_4096_numkeep: excess +18.7 points; session-level 95% CI +16.0 to +21.3 (5 vs 5 sessions); turn-level 95% CI +10.0 to +27.3 (150 vs 150 task turns) (10000 resamples each, seed 20261014; the paper uses the session-level CI)

`x2-evalset-tasks-real-v2-mistral-small32-wrong-rate` (PENDING, n n/a)

not computed: data file(s) not synced yet: results/x2_evalset_tasks_real_v2_mistral_small32.jsonl

`x2-evalset-tasks-real-v2-qwen3-8b-wrong-rate` (VERIFIED, n 0 task turns)

run finished False; qwen3:8b: refused (qwen3:8b: task tool called correctly on 61/81 baseline turns (< 90%, task-tool gate); qwen3:8b: task end-state gate failed (65/90 < 90%); qwen3:8b: strict preflight, gates not passed: ['baseline', 'task_tool', 'task_end_state']) (bootstrap 10000 resamples, unit session, stratified by tier, seed 20261014)

`x2-evalset-tasks-real-v2-wrong-rate` (VERIFIED, n 750 task turns)

run finished True; llama3.1:8b: refused (llama3.1:8b: negative control failed (canary misses 26 at num_ctx 131072); llama3.1:8b: not in this plan's model list ['qwen3:14b']; llama3.1:8b: task end-state gate failed (72/90 < 90%)) / qwen3:14b: valid True; ollama_default: wrong 17/150 (11.3%) (silent 17, surfaced 0), loaded context [40960]; ollama_ctx_16384: wrong 19/150 (12.7%) (silent 19, surfaced 0), loaded context [16384]; ollama_ctx_8192: wrong 53/150 (35.3%) (silent 53, surfaced 0), loaded context [8192]; ollama_ctx_4096: wrong 63/150 (42.0%) (silent 63, surfaced 0), loaded context [4096]; ollama_ctx_4096_numkeep: wrong 45/150 (30.0%) (silent 45, surfaced 0), loaded context [4096], num_keep per seed {20260901: 3369, 20260902: 3369, 20260903: 3364, 20260904: 3369, 20260905: 3367}; ollama_ctx_16384 excess over default +1.3 points (95% CI -1.3 to +4.0, 5 vs 5 sessions); reading: no excess shown; ollama_ctx_8192 excess over default +24.0 points (95% CI +18.7 to +29.3, 5 vs 5 sessions); reading: ollama_ctx_8192 raises the wrong-task rate; ollama_ctx_4096 excess over default +30.7 points (95% CI +26.0 to +34.7, 5 vs 5 sessions); reading: ollama_ctx_4096 raises the wrong-task rate; ollama_ctx_4096_numkeep excess over default +18.7 points (95% CI +16.0 to +21.3, 5 vs 5 sessions); reading: ollama_ctx_4096_numkeep raises the wrong-task rate (bootstrap 10000 resamples, unit session, stratified by tier, seed 20261014)

`x2-scale-real-default-tier` (VERIFIED, n 3 sessions)

llama3.3:70b: no default-tier session / qwen3:32b: 3 default-tier sessions completed (seed 20260901: 40 turns, 6.18 h, loaded [40960], first over window 24; seed 20260902: 40 turns, 6.01 h, loaded [40960], first over window 25; seed 20260903: 40 turns, 4.66 h, loaded [40960], first over window 26)

`x2-v3-usability` (VERIFIED, n 121)

per-call timeout 900 s (harness DEFAULT_CALL_TIMEOUT_S); timeout_latency/done (share) [context tiers]: llama3.1:8b/llama_server 40/340 (11.8%) [longdoc_120000 x10, longdoc_64000 x10, longdoc_80000 x10, longdoc_96000 x10]; llama3.1:8b/ollama_default 10/340 (2.9%) [longdoc_120000 x10]; qwen3-14b/llama_server 0/340 (0.0%); qwen3-14b/ollama_default 0/340 (0.0%); qwen3-30b-a3b/llama_server 10/340 (2.9%) [longdoc_120000 x10]; qwen3-30b-a3b/ollama_default 0/340 (0.0%); qwen3-32b/llama_server 11/100 (11.0%) [longdoc_32000 x10, trace_mix_24000 x1]; qwen3-32b/ollama_default 0/100 (0.0%); qwen3-4b-2507/llama_server 40/340 (11.8%) [longdoc_120000 x10, longdoc_64000 x10, longdoc_80000 x10, longdoc_96000 x10]; qwen3-4b-2507/ollama_default 10/340 (2.9%) [longdoc_120000 x10]; qwen3-8b/llama_server 0/340 (0.0%); qwen3-8b/ollama_default 0/340 (0.0%)